

Tanvir Singh

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Education

Macquarie University

Bachelor of Information Technology, Software Technology major - WAM 75+

Expected Dec 2026

North Ryde, NSW

- Coursework: Data Structures & Algorithms, Object-Oriented Programming, Systems Programming, Databases, Networking, Distributed Systems, Mobile App Development.

Experience

Software Developer & Documentation Lead

Waranara Library Exhibition, Industry Placement

Feb 2026 – Jun 2026

Sydney, NSW

- Built C# gameplay and UI systems for an interactive Unity exhibit over 12 weekly Agile sprints, delivered a production build praised by the client.
- Led documentation for a five-person team, producing developer and user guides to help the client operate and manage the exhibit independently.

Co-Founder & Technical Operations Lead

Precision Cricket Academy

Mar 2025 – Present

Schofields, NSW

- Built a digital scheduling and training platform for 30+ active players, replacing spreadsheet tracking and saving 3-5 hours of weekly administration.
- Support day-to-day administrative operations coordinating player schedules, payments, training programs, and parent communication.

Academic Tutor (Mathematics & STEM)

Kumon Education

Apr 2019 – Jun 2025

The Hills District, NSW

- Supported 20+ students through individualised tutoring and adaptive teaching, using progress tracking to target learning gaps.

Projects

Sports Academy Platform (In Progress)

React, TypeScript, FastAPI, PostgreSQL, Docker

[GitHub](#)

- Building a full-stack platform for tracking sessions, bookings, and attendance at Precision Cricket Academy, using a Docker-based environment, CI pipeline, and testing with pytest, Vitest, and Playwright.

QuestShaper Multiplayer Game Server

Java, REST APIs, Docker, AWS, Terraform

[GitHub](#)

- Led backend development on a three-person team, building REST endpoints for authentication, game state, inventory, and player actions.
- Automated deployment on push using GitHub Actions, Docker, Terraform, and AWS EC2, replacing manual server restarts and redeployment steps.

Distributed Systems Job Scheduler

Python, TCP sockets, XML

[GitHub](#)

- Co-designed a client-server scheduler over TCP sockets, which reduced average job turnaround time by over 99% versus a baseline in tests of up to 10,000 jobs.
- Evaluated utilisation, cost, and turnaround time to explain the final scheduling approach.

DormDeals – Campus Marketplace App

Flutter, Dart, Firebase, Google Maps APIs

[GitHub](#)

- Independently built a cross-platform marketplace app spanning UI, authentication, database, and backend integration.
- Added Firebase Authentication, real-time Firestore listings, Provider state management, and map-based features, and resolved a post-launch security-rules misconfiguration using network logging.

Technical Skills

Languages: Python, Java, C/C++, C#, Dart, TypeScript, SQL

Cloud & DevOps: AWS EC2, Docker, Terraform, GitHub Actions (CI/CD)

Frameworks & Databases: React, FastAPI, Flutter, Firebase, PostgreSQL, MySQL

Tools & Testing: Git, JUnit, pytest, Vitest, Playwright, Linux, Microsoft Office Suite

Certifications: AWS Certified Cloud Practitioner, HackerRank SWE, SQL, REST API, RCA & CAPA (UNIDO)